

1. What is CSS Box Model?

The CSS Box Model describes how every HTML element is represented as a rectangular box. It consists of four parts:

1. Content: The actual text or image.
2. Padding: Space between content and border.
3. Border: A line surrounding the padding and content.
4. Margin: Space outside the border.

```
div {  
  width: 200px;  
  padding: 10px;  
  border: 2px solid black;  
  margin: 20px;  
}
```

2. What is media query? Explain with a sample code.

A Media Query is used in CSS to make a website responsive for different screen sizes, such as mobile, tablet, and desktop.

```
@media (max-width: 600px) {  
  body {  
    background-color: lightblue;  
  }  
}
```

3. What are the features of Bootstrap?

- **Mobile-First Responsive Grid:** Uses a flexible 12-column system to adapt layouts to various screen sizes.
- **Pre-styled UI Components:** Includes ready-to-use buttons, navbars, modals, cards, and forms right out of the box.
- **Utility Classes:** Provides hundreds of quick helper classes for margin, padding, colors, text alignment, and flexbox behavior without writing custom CSS.
- **Cross-Browser Consistency:** Handles normalizations so layouts look consistent across Chrome, Safari, Firefox, and Edge.
- **Interactive JS Plugins:** Offers built-in JavaScript functionality for accordions, dropdowns, tooltips, and carousels.

4. Write an example of basic grid structure in Bootstrap?

Bootstrap uses a 12-column grid system to arrange content on a webpage.

```
<div class="container">
  <div class="row">
    <div class="col-md-4">Column 1</div>
    <div class="col-md-4">Column 2</div>
    <div class="col-md-4">Column 3</div>
  </div>
</div>
```

- **container:** Holds the content.
- **row:** Creates a horizontal row.
- **col-md-4:** Each column takes 4 out of 12 grid columns.
- **Total:** $4 + 4 + 4 = 12$ columns.

5. What is CSS Selectors?

CSS selectors are patterns used to target specific HTML elements so you can style them. They range from simple tag names to complex relationships between nodes.

- **Element Selector:** Targets HTML tags directly (`p { color: red; }`).
- **Class Selector:** Targets elements with a specific class attribute using a dot (`.btn { display: inline-block; }`).
- **ID Selector:** Targets a single element with a unique ID using a hash (`#main-header { font-weight: bold; }`).
- **Universal Selector:** Matches every element on the page (`* { box-sizing: border-box; }`).
- **Attribute Selector:** Selects elements based on their HTML attributes (`input[type="text"] { border: 1px solid #ccc; }`).

6. What are CSS backgrounds, list the properties?

CSS background properties allow you to set visual effects on the background layer of an element, including plain colors, images, and gradients.

Main Properties:

- **background-color:** Defines the solid color behind an element.
- **background-image:** Specifies one or more background image URLs or gradients.
- **background-repeat:** Controls whether/how the background image tiles (`repeat`, `no-repeat`, `repeat-x`, `repeat-y`).
- **background-position:** Sets the starting position of the background image (e.g., `center top`, `50% 50%`).
- **background-size:** Scales the background image (`cover`, `contain`, or specific lengths/percentages).
- **background-attachment:** Determines if a background image scrolls with the page or stays fixed (`scroll`, `fixed`).

- **background:** The shorthand property to combine all the above in one declaration.

7. How can we overlap elements in CSS?

You can overlap elements in CSS using several straightforward approaches depending on the layout structure:

1. **Absolute/Relative Positioning:** Set the parent element to **position: relative;** and the child element to **position: absolute;**. Adjust **top**, **bottom**, **left**, or **right** to shift the element over another.
2. **Negative Margins:** Apply negative spacing (e.g., **margin-top: -30px;**) to pull an element over an adjacent item in normal document flow.
3. **CSS Grid:** Place multiple child elements into the exact same grid row and column coordinates:

```
.grid-container { display: grid; }
```

```
.item1, .item2 { grid-column: 1; grid-row: 1; }
```

4. **Z-Index Control:** For any overlapped elements, use **z-index** (e.g., **z-index: 10;**) to control which element layers on top of the other.

8. What are the different positioning properties in CSS?

- **static:** Default value. Elements follow the normal document layout flow. **top**, **bottom**, **left**, **right**, and **z-index** have no effect.
- **relative:** Positioned relative to its normal position in the document flow without affecting surrounding elements.
- **absolute:** Positioned relative to its nearest ancestor that has a positioning value other than **static**. It is removed completely from the normal document flow.

- **fixed:** Positioned relative to the browser viewport. It stays pinned in place even when the user scrolls.
- **sticky:** Toggles between **relative** and **fixed** depending on the user's scroll position (stuck relative to a set threshold like **top: 0**).

9. What is ECMAScript?

ECMAScript is a standard specification that defines the features and rules of scripting languages such as JavaScript.

- JavaScript is an implementation of the ECMAScript standard.
- It defines variables, functions, loops, objects, and other language features.

10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead zone?

Difference between var, let, and const

Feature	var	let	const
Scope	Function scope	Block scope	Block scope
Redeclaration	Allowed	Not allowed in same scope	Not allowed in same scope
Reassignment	Allowed	Allowed	Not allowed
Hoisting	Hoisted, initialized as undefined	Hoisted, but uninitialized	Hoisted, but uninitialized